

CH365 Chemical Engineering Thermodynamics

Lesson 2 Fundamentals 2

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Work

When a force acts over a distance, work is force times displacement:

force is F and displacement is dl

Eq. 1.2 $dW = F dl$ positive (+) if F and dl are in the same direction
negative (-) if F and dl are in the opposite direction

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Take Notes!

Energy and Work Overview

Slide 3

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Take Notes!

Heat

“Flows” from region of higher T to region of lower T

Take Notes Here!

Temperature difference is the “driving force” for the flow of energy as heat

Take Notes Here!

The driving force analogy comes from physics:

- voltage difference drives current flow in an electrical circuit
- gravitational potential drives free fall of an object
- pressure difference drives fluid flow in a horizontal pipe
- concentration difference drives molecular diffusion

Heat is transferred between the system and its surroundings.

Take Notes Here!

1 calorie raises the temperature of 1 gram of water 1 deg C

1 Btu raises the temperature of 1 lb_m of water 1 deg F

Take Notes Here!

Questions?

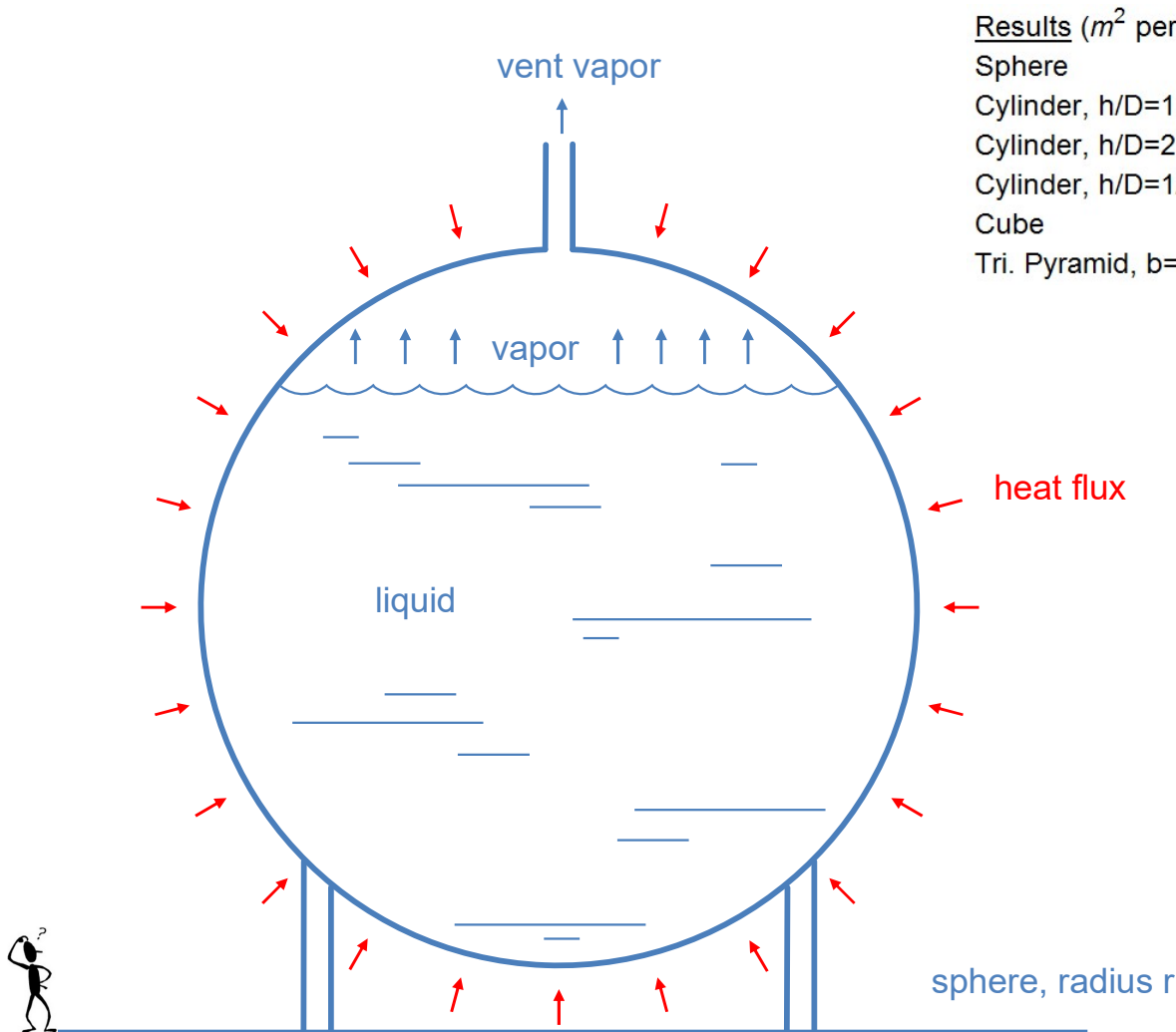
Lesson 2 Problems

Problem 1.11

Liquids that boil at relatively low temperatures are often stored as liquids under their vapor pressures, which at ambient temperature can be quite large. Thus, n-butane stored as a liquid/vapor system is at a pressure of 2.581 bar for a temperature of 300 K. Large-scale storage ($>50\text{m}^3$) of this kind is sometimes done in spherical tanks. Suggest two reasons why.

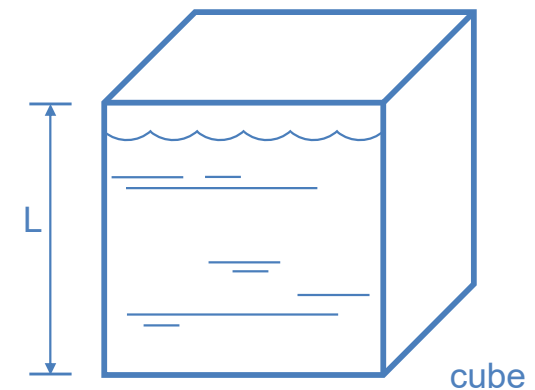
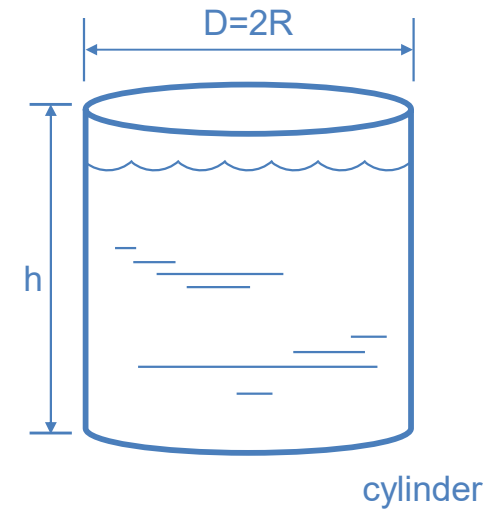
Illustration for Problem 1.11 for Spherical, Cylindrical, and Cubical Tank

Slide 8



Results (m^2 per $50 m^3$):

Sphere	65.6348
Cylinder, $h/D=1$	75.1346
Cylinder, $h/D=2$	78.8840
Cylinder, $h/D=1/2$	79.5104
Cube	81.4325
Tri. Pyramid, $b=h$	89.6128



Heat flux is proportional to temperature difference,
Rate of heat transfer, CH485, RRW, Table 1.5 p.16

$$\frac{Q}{a} = U(T_{\text{outside}} - T_{\text{inside}}) \Rightarrow Q = Ua(T_{\text{outside}} - T_{\text{inside}})$$

$$\text{flux} = \frac{Q}{a} = \frac{\text{flow}}{\text{area}} = \frac{J/s}{m^2}$$

If each vessel has a max capacity
of $50 m^3$ of liquid, what is the
area of each vessel?

$$50m^3 = \frac{4}{3}\pi r^3 = 2\pi R^3 = L^3 \quad (h=D)$$

Once r , R , and L are known, area
for each shape can be calculated.